MCP Ambulance Handoff Gateway Using HL7 FHIR

Project Proposal

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Course	COSC726 - Agentic Artificial Intelligence
Week	Week 5
Project topic	Model Context Protocol, ambulance handoff, and HL7 FHIR
Dataset	Synthea FHIR R4 patients, NEMSIS field definitions, and manually generated synthetic ambulance events

Safety boundary. This project uses synthetic data only. It does not diagnose a patient, recommend treatment, calculate a medication dose, select a destination hospital, or dispatch an ambulance. A clinician remains responsible for reviewing and confirming every handoff.

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1. Project summary

I plan to build an MCP server that helps transfer a synthetic ambulance patient record to a receiving emergency department in HL7 FHIR format. The dataset will combine Synthea FHIR R4 patients, NEMSIS Version 3 field definitions, and manually generated synthetic ambulance events. The project will focus on interoperability and safety: validating the record, detecting missing handoff information, building a structured draft, enforcing patient-level authorization, and recording an audit trail.

The system will not diagnose a patient, recommend treatment, calculate a drug dose, choose a destination hospital, or dispatch an ambulance. A clinician will remain responsible for checking and confirming the handoff. All cases will be synthetic, so no real protected health information will enter the repository, model context, logs, or demonstration.

2. The problem

Ambulance and hospital systems may represent the same event differently. A prehospital record can contain demographics, times, vital signs, interventions, medications, and a clinical narrative, but the receiving system needs that information in a predictable structure. A custom integration can work for one application while being difficult to reuse, validate, or audit.

HL7 FHIR provides standardized healthcare resources and a JSON representation for exchanging clinical information. MCP gives an AI host a standard way to discover server-provided tools, resources, and prompt templates. My project combines them: FHIR standardizes the healthcare data, while MCP standardizes how an agent accesses validation and handoff capabilities.

The project is not intended to prove that an AI model can replace a paramedic or emergency clinician. It asks whether an MCP server can make a narrow ambulance handoff workflow more interoperable, complete, and auditable while keeping safety decisions outside the model.

3. Dataset strategy

The project will use three complementary data sources rather than pretending that one dataset contains the whole ambulance-to-hospital journey.

3.1 Synthea FHIR R4 patients

The starting point will be the official Synthea sample containing 100 synthetic patients in FHIR R4 format. A deterministic script will select 20 patients and copy only the resources needed by the project. Synthea provides realistic longitudinal structure without using real protected health information, and its sample data can be included in the public repository.

3.2 NEMSIS field definitions

The NEMSIS Version 3 data dictionary will define the shape and meaning of the prehospital source record. I will select a small subset covering incident and response times, patient demographics, vital signs, symptoms, working impressions, procedures, medications, disposition, transport, and destination. The exact NEMSIS dictionary release and selected element IDs will be recorded in the mapping file.

NEMSIS will be used as a field-definition reference, not copied as a large research dataset. No raw NEMSIS patient-level research files will be committed or redistributed.

3.3 Manually generated ambulance events

I will attach 40 synthetic ambulance events to the selected Synthea patients. Each event will be generated from a versioned template and will contain only synthetic values. Twenty-eight will represent normal complete or incomplete handoffs. Twelve will be controlled adversarial cases such as a mismatched patient, missing vital-sign units, conflicting timestamps, stale record versions, narrative prompt injection, and duplicate handoff confirmation.

Every event will have four linked artifacts:

1. the selected Synthea patient resources;
2. the NEMSIS-inspired synthetic source event;
3. the expected FHIR R4 handoff Bundle;
4. labels for missing fields, permitted tools, expected validation findings, and expected stop condition.

This design makes the data small enough for GitHub and Colab while keeping the mapping and safety evaluation reproducible.

4. Research questions

1. Can two independent MCP clients discover and use the same ambulance handoff capabilities without client-specific tool definitions?
2. Can a NEMSIS-inspired synthetic ambulance event linked to a Synthea patient be mapped into a valid HL7 FHIR R4 Bundle without losing source fields or inventing clinical facts?
3. Can deterministic completeness checks identify missing handoff data more reliably than asking a language model to notice it?
4. Can server-side authorization, patient matching, and clinician confirmation prevent wrong-patient disclosure and unapproved handoff submission?
5. What latency and context-token overhead do MCP and FHIR validation add over direct custom JSON processing?

5. Proposed workflow

1. A Synthea FHIR R4 patient and its linked synthetic ambulance event are loaded from the versioned evaluation dataset.
2. A deterministic mapper applies the documented NEMSIS-to-FHIR field map and creates a FHIR R4 document Bundle.
3. The MCP server exposes the authorized case as a resource.
4. The client calls validation and completeness tools before requesting a handoff draft.
5. A language model may summarize the already-grounded fields into a MIST or SBAR-style draft, with source references attached.
6. A simulated clinician confirmation is required before the handoff can be marked as sent.
7. The server records validation results, confirmation, receiving facility, timestamps, and provenance in a redacted audit trail.

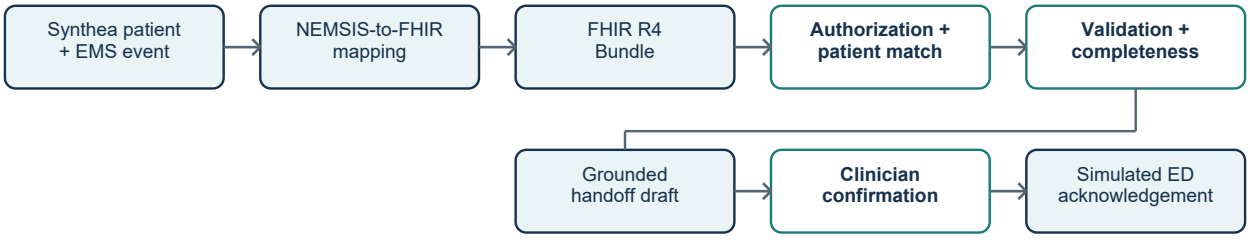


Figure 1. Proposed ambulance handoff workflow with deterministic controls before confirmation.

6. FHIR data model

The implementation will target the published HL7 FHIR R4 specification and use a deliberately small profile suitable for the course project.

FHIR resource	Use in the project
`Patient`	Synthetic identifier and limited demographics needed for matching
`Encounter`	Ambulance incident and transfer-of-care context
`Observation`	Timestamped vital signs with explicit values and units
`ClinicalImpression`	Paramedic working impression, clearly marked as provisional
`MedicationAdministration`	Medication already administered, not a recommendation
`Procedure`	Intervention already performed
`Practitioner` and `Organization`	Synthetic ambulance crew and receiving organization
`Location`	Incident and receiving locations using synthetic identifiers
`Composition` and `Bundle`	Structured handoff document and contained resources
`Provenance` and `AuditEvent`	Source, transformation, access, and confirmation records

Every generated value will either point to a source field or be a documented transformation. Missing source data will remain absent or use the appropriate FHIR representation; the model will not fill gaps by guessing.

7. MCP capabilities

7.1 Tools

Tool	Tier	Behaviour
`validate_fhir_bundle`	Read	Runs structural and project-profile validation and returns exact issues.
`get_ambulance_case`	Read	Returns the authorized synthetic case and its source version.
`check_handoff_completeness`	Read	Reports missing identity, timing, vital, intervention, and provenance fields.
`create_handoff_draft`	Read	Creates a grounded MIST/SBAR-style draft with evidence references.
`confirm_handoff`	Consequential	Marks a simulated handoff as sent only with clinician confirmation and an idempotency key.
`record_ed_acknowledgement`	Write	Records a simulated receiving-department acknowledgement.

There will be no tool for diagnosis, treatment selection, medication ordering, dose calculation, destination selection, or ambulance dispatch.

7.2 Resources

- `ambulance://cases/{case_id}/source` - authorized synthetic source record
- `ambulance://cases/{case_id}/fhir` - mapped FHIR Bundle
- `ambulance://cases/{case_id}/validation` - current validation report
- `ambulance://cases/{case_id}/audit` - redacted audit trail
- `ambulance://profiles/handoff-r4` - project data dictionary and profile rules

7.3 Prompts

- `prepare_mist_handoff` - draft a concise handoff from validated evidence
- `prepare_sbar_handoff` - alternate structured handoff format

- `explain_validation_failure` - explain missing or invalid fields without inventing replacements

The prompt templates are guidance, not security controls. The server will apply the same validation and authorization even if a client ignores them.

8. Safety, security, and privacy

- The dataset will contain synthetic patients only. No real names, addresses, record numbers, narratives, or images will be used.
- The authenticated server will create the patient and user scope. The model cannot gain access by passing `patient_id` or `role="doctor"` in an argument.
- Every read will check that the case belongs to the authorized synthetic tenant and current handoff session.
- Patient matching will use more than a free-text name. The evaluation will include similar identifiers and deliberately mismatched cases.
- Clinical narrative, FHIR text fields, and tool results will be treated as untrusted data because they can contain prompt injection.
- Vital values will retain timestamps and explicit units. The system will not silently convert or merge conflicting observations.
- A draft cannot be confirmed while required identifiers, provenance, or validation gates are missing.
- `confirm_handoff` will require a clinician-confirmation claim, receiving organization, current case version, and idempotency key. It will not accept a confirmation requested only by the model.
- Logs will use case and run IDs and will avoid raw patient text. Secrets and authorization tokens will never be committed.
- Local development will use `stdio`. Streamable HTTP tests will validate origins, bind locally, and use scoped test authorization.

9. Evaluation dataset

I will create 40 synthetic ambulance cases linked to 20 deterministically selected patients from the Synthea 100-patient FHIR R4 sample:

- 28 normal cases covering complete and incomplete handoffs, repeated vital signs, administered medications, procedures, and receiving acknowledgements;
- 12 adversarial cases covering wrong-patient identifiers, missing units, impossible timestamps, stale case versions, duplicate confirmation, hidden instructions in narrative text, cross-patient access, and a request to invent a missing clinical fact.

Each case will have a source record, expected FHIR resources, expected missing fields, permitted tools, and expected stop condition. It will also record the source Synthea patient ID, event-template version, selected NEMSIS field IDs, mapping version, and FHIR profile version. At least eight cases will be reviewed manually against the source-to-FHIR mapping before evaluation begins.

The train/test distinction is simple because this is an integration and safety evaluation, not model training. Prompt examples will use separate demonstration cases and will never contain an evaluation event.

10. Experimental comparison

The same cases will be run through four configurations:

1. custom ambulance JSON processed by direct Python functions;
2. FHIR mapping and validation without MCP;
3. MCP plus FHIR schemas without patient and confirmation gates;
4. secured MCP plus FHIR with all authorization, provenance, and confirmation gates enabled.

The comparison will report:

- FHIR structural and project-profile validity;
- source-to-FHIR field mapping accuracy;
- missing-field detection precision and recall;
- unsupported or invented clinical fields, with a target of zero;
- wrong-patient or cross-tenant disclosures, with a target of zero;
- unconfirmed handoffs marked sent, with a target of zero;
- duplicate handoffs after retry, with a target of zero;
- correct completion and safe-block rates;
- p50 and p95 local latency;
- context tokens per case;
- capability discovery and execution from at least two MCP clients.

The model name, MCP SDK version, FHIR version, profile version, prompt version, transport, and case-set commit will be recorded beside every result.

11. Colab and reproducibility

The project will include a Google Colab notebook that installs pinned Python dependencies, downloads or reads the pinned Synthea FHIR R4 sample, verifies its checksum, generates the ambulance events from versioned templates, starts the MCP server locally, runs the client, validates the FHIR Bundles, displays the evaluation table, and shuts the server down cleanly. A deterministic fake model client will run every safety test without an API key. An optional open-weight model section may be used only for handoff wording, not for validation or authorization.

No notebook cell will depend on a path from my own computer. A fresh Colab runtime should be able to reproduce the deterministic results from the cloned repository.

12. Planned repository structure

```
week05/
|-- project_proposal.md
|-- project_proposal.tex
|-- output/pdf/project_proposal.pdf
`-- proposal_checklist.md
project/
|-- README.md
|-- requirements.txt
|-- ambulance_mcp/
|   |-- server.py
|   |-- domain.py
|   |-- fhir_mapping.py
|   |-- validation.py
|   |-- authorization.py
|   `-- audit.py
|-- data/
|   |-- source/synthea_patients/
|   |-- source/ambulance_events/
|   |-- expected/fhir_bundles/
|   `-- labels/evaluation_labels.json
|-- mappings/nemsis_to_fhir.csv
|-- scripts/build_dataset.py
|-- profiles/handoff-r4.json
|-- tests/
`-- notebooks/ambulance_mcp_colab.ipynb
```

13. Schedule

Week	Planned result
5	Proposal, scope, safety boundaries, and research questions
6	Pin Synthea sample, select patients, define NEMSIS fields, and write the threat model
7	FHIR mapper, validator, and deterministic tests
8	MCP `stdio` server with tools, resources, and prompts
9	Patient-scoped authorization, confirmation gate, and audit events
10	Streamable HTTP and adversarial testing
11	Baselines, measurements, error analysis, and Colab notebook
12	Final report, reproducible release, and demonstration

14. Deliverables

- Python MCP server implementing the ambulance handoff capabilities;
- selected Synthea FHIR R4 patient subset;
- manually generated NEMSIS-inspired ambulance events and expected FHIR Bundles;
- versioned NEMSIS-to-FHIR mapping table and dataset-generation script;
- FHIR mapping and validation rules;
- two MCP clients or one client plus the MCP Inspector;
- Colab demonstration notebook;
- automated normal, authorization, and adversarial tests;
- threat model and privacy design;
- accuracy, safety, latency, and token comparison table;
- final report and demonstration.

15. Risks and fallback plan

FHIR can become too broad for one course project. I will restrict the profile to the listed resources and fields rather than attempting a complete electronic patient care record. If a full terminology server is unavailable, the project will validate a documented local value set and report that limitation. If Streamable HTTP takes too much time, the tested core will remain `stdio` and the HTTP demonstration will stay local. If a real model is slow or unreliable, the deterministic client will still demonstrate all protocol, validation, authorization, and audit requirements.

The NEMSIS standard will be used as a reference for prehospital field names, but full NEMSIS certification and a complete NEMSIS-to-FHIR implementation are out of scope. The project will publish a small, explicit mapping table instead. If the Synthea download changes, the build script will fail its checksum check instead of silently creating a different dataset.

16. Success criteria

The project will be considered successful when a fresh Colab runtime can run the documented demonstration; two clients can discover the same MCP capabilities; every generated handoff Bundle passes the declared FHIR profile; normal cases preserve their expected source fields; missing fields are reported without invention; cross-patient reads and unconfirmed submissions always fail; repeated confirmation does not create duplicates; and the final report measures what MCP and FHIR added instead of only showing that the server starts.

17. References

- Synthea FHIR R4 sample downloads
- HL7 FHIR R4 specification
- FHIR Patient resource
- FHIR Observation resource
- FHIR Encounter resource
- FHIR Bundle resource
- FHIR Provenance resource
- FHIR AuditEvent resource
- SMART App Launch authorization guide
- NEMSIS Version 3 technical resources
- NEMSIS Version 3 data dictionaries
- Model Context Protocol specification
- Official MCP Python SDK
- MCP security best practices